ANSI/IES LM-98-24

Illuminating Engineering Society

APPROVED METHOD: MEASURING IN-SITU TEMPERATURE OF SOLID-STATE LIGHTING COMPONENTS IN LAMPS AND LUMINAIRES

AN AMERICAN NATIONAL STANDARD

Approved American National Standard ANSI

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has been approved by IES.

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should be directed to IES.

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1.0 Introduction and Scope

1.1 Introduction

Solid state lighting (SSL) products have a temperature dependence that affects performance. In the past decade, the industry has been measuring in-situ temperature characteristics of its assemblies or major components installed in integrated and non-integrated lamps and luminaires. This type of measurement is performed to obtain information on how these components behave in each individual lamp or luminaire and how that product's operating temperature affects its lifetime performance, including lumen maintenance.

A consistent in-situ temperature measurement method would support the understanding of how the SSL product would behave in a real-world environment, so that the operating temperature could then be utilized in more reliable projections for the lumen maintenance and driver lifetime.

Product characteristics obtained using this procedure are measured under controlled conditions that may allow direct comparison of results gathered at different laboratories using this method.

This document is intended to provide a standalone test method for laboratories interested in measuring performance of SSL lamps and luminaires without addressing safety testing. The intention is that this method should align with safety and performance measurements such that they could be performed during the same test or with the same test setup, to allow for minimal impact to testing laboratories or minimal increased testing burden on manufacturers.

1.2 Scope

The document defines a method of measurement of the in-situ temperature of SSL components installed in integrated and non-integrated SSL lamps and luminaires. The method describes the procedures to be followed and the precautions to be observed in obtaining and reproducing in-situ temperature of SSL component measurements under standard operating conditions.

2.0 Normative References

This Approved Method is intended to be used in conjunction with the publications described in Sections 2.1 through 2.4; the latest edition of the publication shall apply.

2.1 ANSI/IES LM-80-21

Illuminating Engineering Society. Approved Method: Measuring Maintenance of Light Output Characteristics of Solid-State Light Sources. New York: IES; 2021.

2.2 ANSI/IES LS-1-22

Illuminating Engineering Society. Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2022. Online: www.ies.org/standards/definitions/. (Accessed 2024 Mar 29).

2.3 ASTM Standard E230/E230M-17

ASTM International. ASTM E230/E230M-17, Standard Specification and Temperature-Electromotive Force (EMF) Tables for Standardized Thermocouples. West Conshohocken, PA: ASTM International; 2017.

2.4 Conditionally Normative References

The product-type-specific standards marked "conditionally normative" are normative depending mounted roadway luminaire is being measured, then UL 1598 is normative, but UL 1993 and UL 153 are not normative.

2.4.1 UL 153. Underwriters Laboratory. UL 153, 13th ed., UL Standard for Safety – Portable Electric Luminaires. Northbrook, IL: UL; 2022 Sep 26.

2.4.2 UL 1598. Underwriters Laboratory. UL 1598, 5th ed., UL Standard for Safety – Luminaires. Northbrook, IL: UL; 2024 Jan 31.

2.4.3 UL 1993. Underwriters Laboratory. UL 1993, 6th ed., UL Standard for Safety – Self-Ballasted Lamps and Lamp Adapters. Northbrook, IL: UL; 2023 Feb 17.

2.4.4 UL 2108. Underwriters Laboratory. UL 2108, 2nd ed., UL Standard for Safety – Low Voltage Lighting Systems. Northbrook, IL: UL; 2015 Dec 7.

2.4.5 UL 8800. Underwriters Laboratory. UL 8800, 1st ed., UL Standard for Safety – Horticultural Lighting Equipment and Systems. Northbrook, IL: UL; 2023 Sep 14.

3.0 Definitions

3.1 ambient temperature (T_A)

The temperature of the air surrounding the device under test (DUT) during the test.

3.2 device under test (DUT)

The integrated SSL lamp, non-integrated SSL lamp, or SSL luminaire that is undergoing the test.

3.3 drive current

The current applied at the SSL component of the DUT to achieve intended operation. Expressed in amperes.

3.4 driver case temperature measurement point (TMP_C)

The SSL driver manufacturer's designated location on the driver case where the temperature may be associated with the DUT driver's performance.

3.5 driver case TMP_C temperature (T_C)

The temperature measured at the TMP_C for the SSL driver.)

3.6 SSL component temperature measurement point (TMP_S)

The SSL (light source) component manufacturer's designated location on the SSL component, such as an LED package, array, or module or a nearby point external to the LED package, in which the temperature associated with DUT's performance is measured. (*Note:* The TMPs is the same as the TMP described in the ANSI/IES LM-80-21 report for that LED package, array, or module to be utilized for life projection calculations.)

3.7 SSL component TMP_S temperature (T_S)

The temperature measured at the TMP_S for the LED package, array, or module.

4.0 Physical and Environmental Conditions

4.1 General

It is recommended laboratory practice that the storage and testing of DUTs be undertaken in a relatively clean environment. Prior to operation, DUTs should be cleaned to eliminate handling marks, and the manufacturer's handling instructions should be observed, e.g., electro-static discharge (ESD) instructions.

4.2 Humidity

During the test, the relative humidity (RH) should be maintained either at less than 65% or at another pre-defined RH level (for specific applications).

4.3 Ambient Temperature and Air Movement

During the maintenance test, the impact of airflow (e.g., forced or natural convection) on the temperature of the TMPs should be minimized. Ambient temperature, T_A, shall be maintained at a temperature of 25 ± 5 °C, unless specified otherwise for application specific requirements. T_A monitoring shall adhere to the product-type-specific conditionally normative reference standard (refer to **Section 2.4**).

4.4 Temperature Measurement Equipment or System

The temperature measurement equipment or system shall have an expanded uncertainty (k = 2) of less than 2.5 °C. If thermocouples are used, the wire shall comply with ASTM E230 Table 1 Special Limits.

5.0 Electrical Conditions

5.1 DC Voltage Regulation

It is recommended that DC voltage regulation conform with ANSI/IES LM-79-19, Section 5.1.3.[1] If the power source is not regulated per ANSI/IES LM 79-19, Section 5.1.3, then it shall conform to the requirements of the applicable conditionally normative standard (see **Section 2.4**), and that standard shall be reported.

5.2 AC Voltage Regulation

It is recommended that AC voltage regulation conform with ANSI/IES LM-79-19, Section 5.1.2.[1] If the power source is not regulated per ANSI/IES LM 79-19, Section 5.1.2, then it shall conform to the requirements of the applicable conditionally normative standard (see **Section 2.4**), and that standard shall be reported.

5.3 Monitoring of Driver Input and Output

The DUT shall be operated using the designed and supplied driver, if applicable. Power characteristics shall be monitored on the input to the driver during the test. The output power characteristics of the driver should be monitored for use in establishing drive current at the SSL component level, if feasible.

5.4 Drive Current

If output power characteristics of the driver are monitored, or if an SSL component is isolated and current is measured at the SSL component, the drive current at the component can be determined and used for future analysis (e.g., ANSI/IES TM-21-21[2] calculations). (See **Annex B** for examples of analyzing circuits and deriving drive current.)

6.0 Preparation for Measurements

6.1 In-Situ Primary Application

The DUT shall be reviewed for design intent and for the in-situ primary application. Once this in-situ primary application is determined, then the worst-case thermal condition for that application shall be used. The DUT's mounting orientation and the safety standard used for this test shall be referenced in the report, along with photographic evidence of installation.

6.1.1 Portable Lamps and Luminaires (Conditionally Normative Reference: UL 153). A portable lamp or luminaire shall be installed or supported to simulate intended use, in accordance with the manufacturer's instructions. Where more than one method may be used, the device shall be installed or supported to allow recording of the maximum temperature that can be encountered under the intended uses. (See **Table 6-1**.)

Table 6-1. Applicable Reference in UL 153

Intended Use Case	UL 153 Reference
Freestanding and surface mount units	Sections 143 – 145
Portable cabinet lights	General Sections 143 and 144, Specific Test Requirements Section 146
Work lights	General Sections 143 and 144, Specific Test Requirements Section 147

6.1.2 Lamps (Conditional Normative Reference: UL 1993). A lamp shall be installed or supported to simulate intended use, in accordance with the manufacturer's instructions. Where more than one method may be used, the device shall be installed or supported to allow recording of the maximum temperature that can be encountered under the intended uses. (See **Table 6-2**.)

Table 6-2. Applicable Reference in UL 1993

Intended Use Case	UL 1993 Reference
Rated "Not for Enclosed or Recessed Fixtures"	Open to ambient; position base up or base down unless position restricted
Rated "Not for Enclosed Fixtures"	Test apparatus per Figure 9.1, with bottom cylinder aperture open and positioned base up unless position restricted (if lamp will not fit per Figure 9.1, adjust per Section 9.5.2)
Rated "No Restrictions"	Test apparatus per Figure 9.1, with bottom cylinder aperture closed and positioned base up unless position restricted (if lamp will not fit per Figure 9.1, adjust per Section 9.5.2)
Rated "For Open Ceiling Mounted Luminaire Only"	Test apparatus per Figure 9.2, with lamp base up

(continued next page)

Table 6-2. Applicable Reference in UL 1993 *(continued)*

Intended Use Case	UL 1993 Reference
Rated "For Specific Ceiling Mounted Luminaire Only"	Test apparatus per Figure 9.2, installed in specified luminaire
Horizontally mounted lamps	Installed in representative luminaire that will result in the most onerous self-heating
Linear lamps	Installed in representative luminaire that will result in the most onerous self-heating, or in test apparatus per Figure SA9.1

6.1.3 Luminaires (Conditional Normative Reference: UL 1598 and UL 8800). A luminaire shall be installed or supported to simulate intended use, in accordance with the manufacturer's instructions. Where more than one method may be used, the device shall be installed or supported to allow recording of the maximum temperature that can be encountered under the intended uses. (See **Table 6-3**.)

6.1.4 Luminaires for Low Voltage Lighting Systems (Conditionally Normative Reference: UL 2108). A luminaire shall be installed or supported to simulate intended use, in accordance with the manufacturer's instructions. Where more than one method may be used, the device shall be installed or supported to allow recording of the maximum temperature that can be encountered under the intended uses. (See **Table 6-4**.)

6.1.5 Other Mounting. Any mounting configuration not described in **Sections 6.1.1** through **6.1.4** (e.g., hazardous location applications) shall be documented with justification for mounting and for what application it represents. The DUT should be installed during in-situ testing per an appropriate document (e.g., a safety standard) for the type and intended primary application of the DUT.

6.2 Thermocouple Location Selection for Solid State Components

The DUT shall be evaluated for the thermocouple location according to the following methods:

Table 6-3. Applicable Reference in UL 1598 and UL 8800

Intended Use Case	UL 1598 and UL 8800 Reference
Recessed luminaires – IC rated (insulation contact)	Test apparatus per Clause 19.15
Recessed luminaires – non-IC rated (non-insulation contact)	Test apparatus per Clause 19.13
Surface mounted – ceiling	Test apparatus per Clause 19.10
Surface mounted – wall	Test apparatus per Clause 19.11
Pole or post mounted	Test apparatus per Clause 19.1.4

Table 6-4. Applicable Reference in UL 2108

Intended Use Case	Reference
Surface mounted luminaire	Test apparatus per clause 34.3.1.1
Ceiling mounted luminaire	Test apparatus per clause 34.3.1.2
Wall mounted luminaire	Test apparatus per clause 34.3.1.4
Recessed mounted luminaire marked "Type Non IC Recessed"	Test apparatus per clause 34.3.2.1
Recessed mounted luminaire marked "Type IC Recessed"	Test apparatus per clause 34.3.2.2
Cabinet mounted	Test apparatus per clause 34.3.3.1
Under-cabinet or shelf mounted	Test apparatus per clause 34.3.4.1
Storage area of clothes closet	Test apparatus per clause 60.4

- Thermal imaging assessment of the SSL components may be used to find the hottest relative component(s) that would be used to evaluate the DUT. **Figures 6-1** and **6-2** show examples.

  Note: When utilizing thermal imaging, it is important to consider the emissivity coefficient of the materials within the image. For relative measurements examining LEDs of the same type, there should not be any issues with determining the highest-temperature LED. When examining relative measurements for boards with more than one type of LED (e.g., a board containing phosphor-converted LEDs and red LEDs), emissivity coefficient differences are typically considered when determining the highest temperature LED for each type.
- Alternatively, the schematic and mechanical layout (e.g., board location, heat sink) may be used to determine the probable hottest component locations for thermocouple placement.
- For the purposes of an ANSI/IES TM-21-21 life projection, only the T_S defined in the ANSI/IES LM-80-21 report shall be used. Optionally, a secondary TMP_S Location (e.g., on a board or array) may be measured.

A minimum of one SSL component location shall be required, but three or more are recommended, if applicable. The method of component selection shall be documented in the report.

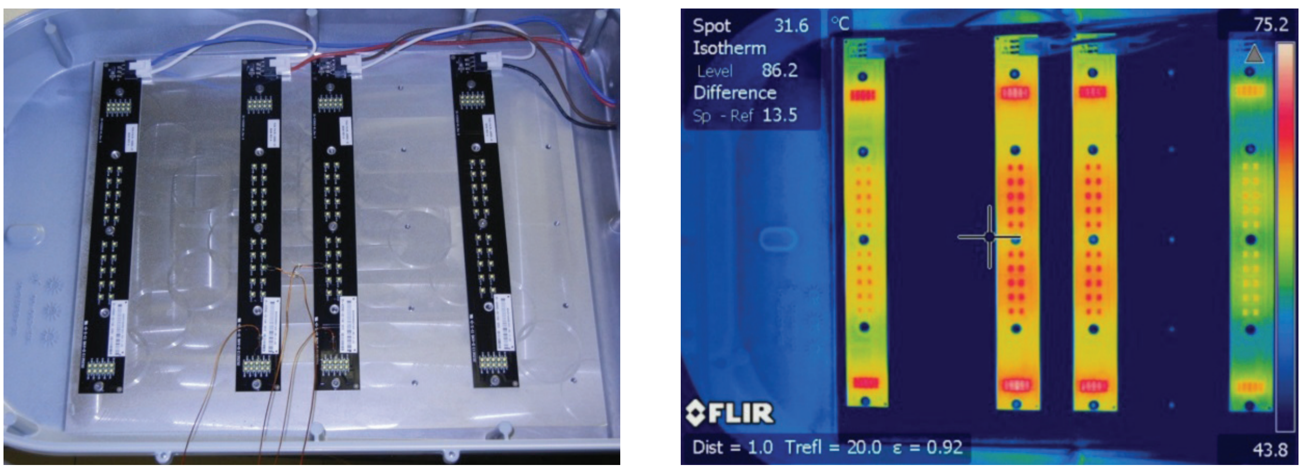

Figure 6-1. ***Left:*** **Typical luminaire assembly with printed circuit board assemblies (PCBA) exposed for IR thermography or non-contact IR thermometer measurement.** ***Right:*** **Typical IR thermograph from powered PCBAs in a luminaire assembly.**

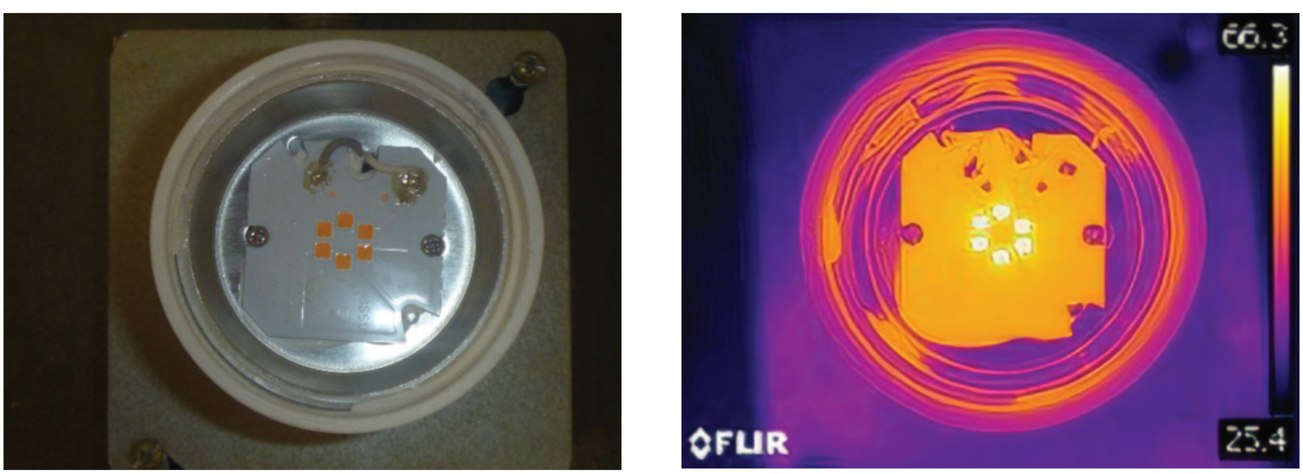

Figure 6-2. ***Left:*** **Typical lamp assembly with PCBAs exposed for IR thermography or non-contact IR thermometer measurement.** ***Right:*** **Typical IR thermograph from powered PCBAs in a lamp assembly.**

Some SSL products may be equipped with multiple models of SSL components (for example, red and blue LEDs may be installed in a horticultural lighting product). In this case, the hottest of each of the distinct SSL component models shall be measured. (Note that each component model in the same product may have different TMP_S locations.)

Thermocouple location for the solid state component shall correspond with the reported TMPS temperature for the SSL component identified in the lumen maintenance document (i.e., the ANSI/IES LM-80-21 report) for the SSL component.

6.3 Thermocouple Location Selection for Driver

This section is applicable if the driver TMP_C is required for the test.

The DUT shall be evaluated for the appropriate thermocouple location using one of the following methods:

- Use the TMP_C location specified by the driver manufacturer, as stated on the driver specification sheet. An example is shown in **Figure 6-3**.
- One of the following methods may be suitable to determine the highest temperature location on the driver. It should be noted, however, that these might not be considered acceptable means of determining the thermocouple location by regulatory or certification bodies.
  1. Conduct a thermal imaging assessment of the SSL components, to find the hottest relative component(s).
  2. Use the manufacturer's schematic and mechanical layout (e.g., electrolytic capacitor, critical temperature component of driver) to determine the probable hottest component(s). An example is shown in **Figure 6-4**, where the electrolytic capacitor C23 was identified and selected as the thermocouple location.

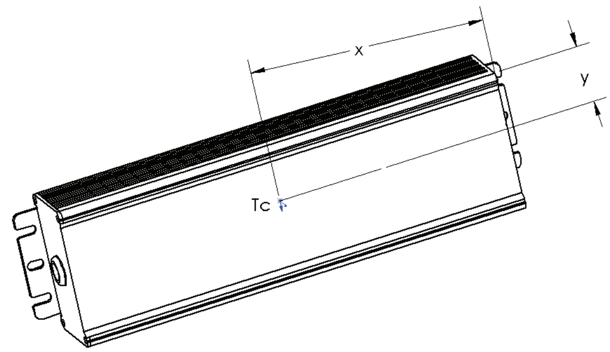

Figure 6-3. Typical enclosed LED Driver with T_C location identified where thermal performance can be measured.

The thermocouple location for the driver(s) shall be documented in the report.

7.0 Thermal and Electrical Measurement Methods and Procedures

Once the DUT is prepared per the conditionally normative reference standard (see **Section 2.4**), then the following procedures shall be followed:

7.1 DUT Thermal Stability

The defined voltage shall be applied to the LED driver.

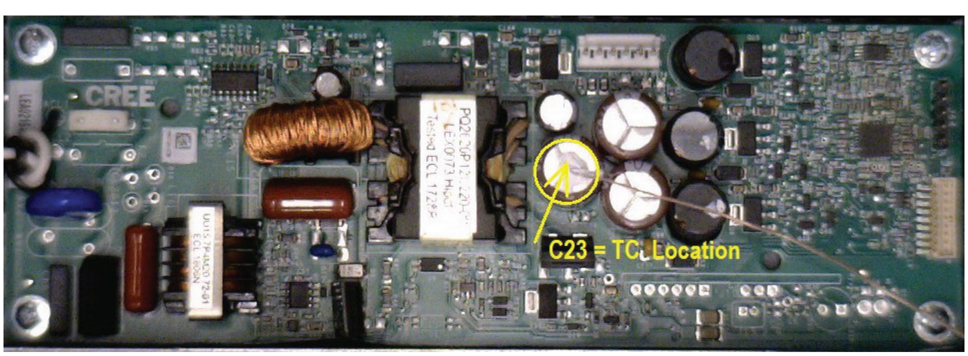

Figure 6-4. Typical open-frame LED driver with T_C location identified where thermal performance can be measured. The thermocouple may be placed directly over the identified component on the outside of the driver enclosure.

It is recommended that the voltage used during the test be the voltage for the lowest driver efficiency, which is typically the highest current for the driver, such as 120 V for a 120V-277V universal driver, and which may yield the highest in-situ driver TMP_C. The DUT shall be monitored until thermal stability is reached, which is when the measured input power and ΔT (T_S or T_C minus T_A) vary less than 0.5% (between maximum and minimum) over three consecutive measurements taken at of 10 minutes apart over a 20-minute period, with the divisor being the last of these measurements chronologically.

Example: If the input power, two LED T_S values, and the driver case T_C are measured, then all four measurements shall meet the defined criteria for the DUT to be deemed thermally stable.

7.2 DUT In-Situ Temperature

After thermal stability is reached per **Section 7.1**, the following values shall be measured and recorded:

- Input power, in watts
- LED driver output current (if measured), in amperes
- Temperature at TMP_C and TMP_S in degrees Celsius

The duration of the test may be extended to meet the conditionally normative reference requirements (refer to **Section 2.4**) for thermal stability, if required.

7.3 Data Normalization

Ambient temperature deviation above T_A of -5 °C or below T_A of +5°C shall be respectively subtracted from or added to the actual measured temperature values on the DUT. Examples are shown in **Table 7-1**.

Table Table 7-1. Examples of Recorded and Normalized Values

Recorded Value	Normalized Value
T_A = 24.7 °C	T_A = 25.0 °C
T_C = 57.5 °C	T_C = 57.8 °C

1. The DUT shall be placed in an elevated ambient temperature environment using a source of heated air to provide the elevated T_A. The maximum airflow past the DUT shall be less than 9.1 m/min (30 ft/min) unless otherwise specified in the conditionally normative reference. Measurements shall be normalized following **Section 7.3**
2. The DUT shall be measured at a T_A of 25 ± 5 °C. Then the difference between the measured temperature values T_C and T_S in the above ambient temperature and the target elevated ambient temperature shall be added to the measured temperature values.

7.5 Reduced Ambient Temperature

Tests may be conducted for a reduced ambient temperature following this procedure:

1. The DUT shall be placed in the reduced ambient temperature environment with a source of cooled air providing the reduced ambient temperature value of T_A.
2. The maximum airflow past the luminaire shall be less than 9.1 m/min (30 ft/min). The measured temperature values T_C and T_S shall be normalized following **Section 7.3**.

8.0 Test Report

The report shall list all pertinent data concerning laboratory location, conditions of testing, type of equipment, and items specified in the body of the procedure. The items listed in **Sections 8.1** through **8.3** shall be reported.

8.1 Administrative Information

- Testing agency identification
- Report issue date
- Testing start date
- Testing completion date

8.2 DUT Identification

- DUT submitter's name (if available to laboratory)
- DUT identification, e.g., model number
- Description of DUT with SSL component identified and driver
- In-situ primary application, mounting and orientation

8.3 Test Conditions and Measurements

- Input power and electrical characteristics
- Output power and electrical characteristics (if monitored)
- TMP location(s), including one or more images with sufficient detail that the location can be replicated
- Ambient temperature
- Measured TMP temperature value(s) once thermally stabilized
- Normalized TMP temperature values adjusted from actual ambient temperature to the target nominal temperature
- Drive current at SSL component if measured or calculated
- Length of time to achieve thermal stability

8.4 Test Equipment

- Description of testing equipment and models used (listing of models used is recommended, not required)

9.0 Measurement Uncertainty

A testing laboratory shall have and apply procedures for estimating uncertainty of measurement. To determine the measurement uncertainty, the calibration certificates of the equipment used during testing shall have an indication of the uncertainty. (For application of measurement uncertainty, refer to guide JCGM 100:2008, *Guide to the Expression of Uncertainty in Measurement* (GUM).[3])

Annex A
Thermocouple Location and Attachment Guide

Note: This annex is not part of ANSI/IES LM-98-24 *but is included for information only.*

A.1 Attachment Guide

A.1.1 Thermocouple "Type" Selection. It is important to ensure that the right thermocouple type (e.g., J, K, T, E) has been selected for the application and temperature range of the test. Consideration should be given to the type of thermocouple that is to be used when temperature measurements are to be made in the presence of magnetic fields. Some types of thermocouples, e.g., Type J thermocouples, are magnetic. Others, e.g., Type K thermocouples, are slightly magnetic. Type T thermocouples are nonmagnetic. Type T thermocouples are preferred for applications that require temperature measurement in the presence of magnetic fields.

A.1.2 Attachment Procedure

1. Place the junction of the thermocouple wire firmly against the surface it will be bonded to.
2. While holding the thermocouple wire, apply a small drop of adhesive to the thermocouple wire junction and surface area.
3. Before using an accelerator, ensure that the following items are observed:
   a. Thermocouples must not contact live parts.
   b. A continuous transition of adhesive between the thermocouple and the bonding surface is preferred.
   c. The thermocouple junction should still be in firm contact with the bonding surface.
   d. There should not be any excess adhesive in contact with adjacent surfaces or components.
   e. Remove any excess adhesive with a lint-free wipe or swab moistened with water.
   f. If removal of excess adhesive cannot be done easily, remove the thermocouple, clean the surface as described above and repeat step above.
4. While holding the thermocouple wire steady, spray a small amount of accelerator onto the adhesive and the thermocouple.
5. Continue to hold the thermocouple wire as the adhesive hardens. (Air lightly blown on the surface will allow the adhesive to dry faster.)
6. Inspect the bonded thermocouple point.
7. Ensure that the thermocouple wire is still properly aligned.
8. Ensure that the adhesive has completely dried.
9. Ensure that the thermocouple wire does not short current-carrying parts.

10. Gently pull the thermocouple wire to ensure that the adhesive bond is secure.
11. Recommended: Bundle thermocouple wires together to provide strain relief during setup and testing of DUT.

    Note: Additional information and guidance may be found in IECEE OD-5012.[4]

A.2 Densely Packed Arrays and Thermocouple Location: Troubleshooting TMP Location Issues

This section provides guidance on the selected location and method of applying thermocouples to measure the driver TMP_C and the SSL TMP_S. The examples below are not exhaustive. However, the rationale may be applicable to other similar scenarios. It is important to note that the following is guidance (a recommendation) and not a requirement.

A.2.1 Selecting the Appropriate SSL Component on the Appropriate PCBA. During DUT preparation, choosing the most appropriate LED(s) to measure can be critical and can be the determining factor as to whether a test produces useful information. The most appropriate LED is commonly the LED that will be the hottest. The unlimited design variations of SSL products can make the SSL component selection difficult; a few guidelines are listed below:

a. Refer to **Section 6.2 Thermocouple Location Selection for Solid State Components** for information on utilizing thermal images or the schematic and mechanical layout. Thermal imaging can be an objective measurable method to determine relative temperatures and select the hottest SSL component.
b. Before selecting the SSL component to measure, the DUT may have multiple PCBAs. The PCBA chosen should be the hottest available, which is commonly the PCBA closest to the center of the DUT.
c. When selecting the SSL component to measure, the hottest SSL component is commonly the LED closest to the center of the DUT and closest to other heat sources.

It is important to note that the guidelines above may be mutually exclusive. For example, an SSL component directly beneath a driver may be hotter than the SSL component closest to the center of the DUT, or an SSL component that at the center of the DUT may have a direct heat path to a heatsink and be cooler than an SSL component at the extremity of the DUT. Using thermal imaging to help determine the hottest SSL component is advised.

It is critical that the distance between the edge of the SSL component package and the LED_{TMP} be able to accommodate the thermocouple wire gauge by a factor 2.5 times, to allow for attachment to the location and accurate measurement of the SSL TMP_S. If the distance between the edge of the SSL package and the TMP_S is less than the necessary spacing for the SSL packages within the DUT, thus preventing appropriate thermocouple location, then this methodology cannot be utilized. In this case, it is recommended that the densely packed array be tested per ANSI/IES LM-80 as an array and not as an individual SSL package.

A.2.2 Thermal and Geometric Symmetry. In many cases, conditions of symmetry can be used to reduce the total number of SSL components that require measurement to obtain a useful representation of the T_S conditions in the SSL product. **Figure A-1** shows an example of a typical PCBA with mid-line symmetry in two dimensions. In the case of for which symmetry exists, it is sufficient to attach a thermocouple to a subset of the SSL components, in reflection of the PCBA symmetry.

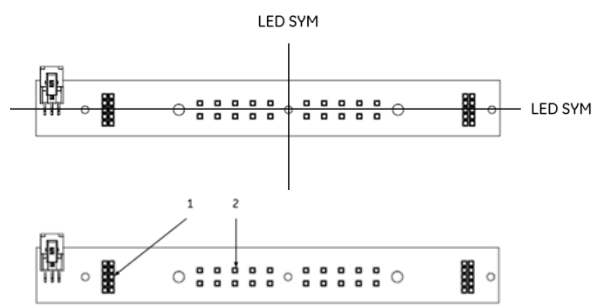

Figure A-1. ***Top:*** **Example of PCBA with mid-line symmetry.** ***Bottom:*** **Example of selected SSL components for thermocouple attachment on a PCBA with four-quadrant symmetry based on thermographic assessment of high-temperature groups.**

A.2.3 Minimizing Measurement Impact for Drivers With Thermal Interface Materials. To help dissipate heat, thermal interface materials (solids, liquids, or gels) may be applied on drivers within DUTs. It is important that extra care be taken to reduce the impact of measurement on the effectiveness of the thermal interface materials. These materials rely on consistent contact between the driver, thermal interface material, and mounting structure. Attaching a thermocouple may cause gaps in that contact.

1. For liquids and gels, discussing the best way to minimize the testing impact on the measurement with someone who is familiar with the DUT design is advised.
2. For solid materials, cutting a path through the thermal interface material from the thermocouple attachment point to the edge of the driver will maintain the most consistent contact between the driver, thermal interface material, and mounting surface.

A.2.4. Thermocouple Placement on Internal Layer of PCBA. In some cases, such as for ANSI/IES TM-21 projections, thermocouples are required to be placed on internal layers of the PCBA due to the testing method used for ANSI/IES LM-80 compliance. In these scenarios, the PCBA is typically designed to accommodate the necessary thermocouple placements. If this is not the case, the PCBAs will need to be modified to allow for appropriate thermocouple placement. Detailed discussions should occur between the testing lab and those knowledgeable about the PCBA to determine the necessary modifications.

Annex B
LED Drive Current

Note: This annex is not part of ANSI/IES LM-98-24 *but is included for information only.*

This annex provides guidance on approaches to identifying methods to measure or estimate the LED drive current relative to thermal applications, such as evaluation of lifetime projections using ANSI/IES TM-21 calculations.

B.1 Basic Assessment

A combination method of measuring and calculating may be used. After the thermal in-situ test is complete, the wiring at the output of the driver may be cut or disconnected and routed in series through a multimeter to measure the output current of the driver. The DUT should be allowed to operate at least five minutes before the driver output current is measured. The DUT should be electrically stable per **Section 7.1 DUT Thermal Stability**. This measured current may then be divided by the number of parallel legs in the circuit, which can be found based on observation, a circuit schematic, or by manufacturer statement.

B.2 Alternative Assessments

Depending on the circuit design, the SSL component drive current may be calculated by measuring the electrical energy through a nearby component (e.g., a voltage drop across a limiting resistor) and applying Ohm's law to the results ($V = I \cdot R$.)

In some integrated designs it may be necessary to cut the metal trace near an LED and solder wires onto the exposed ends to measure the current in series with a multimeter.

The manufacturer may declare the design drive current of the SSL component when it is impractical (i.e., with risk to proper operation of the DUT) to access a measurement location between the driver circuit and the SSL components. The report should clearly state that that a manufacturer-declared rather than a measured SSL drive current was used, due to risk of improper operation of the DUT.

Annex C
Example Guide for Using ANSI/IES LM-98

Note: This annex is not part of ANSI/IES LM-98-24 *but is included for information only.*

This annex provides an example for how a sample luminaire would be evaluated and tested using the method described in ANSI/IES LM-98-24.

C.1 LED Linear Ambient Luminaire In-Situ Temperature Measurement Test

A manufacturer has a 30-W, 4,000-lm, 4-ft linear ambient or strip luminaire.

The DUT selected has a CCT of 3000 K, as it is the lowest CCT offered in that product family and, per the manufacturer, has the highest drive current to achieve the 4,000-lm specified (vs. 4000 K and 5000 K). (*Note:* The DUT under test should be the product that is expected to be generate the most thermal energy compared with any other variation within the product family, is typically operated at the highest drive current, using the least efficient optics, using the smallest housing and/or heat sink, and having the lowest CCT.)

Thermal imaging analysis of the LED board when powered shows that LED "G41" is the hottest LED on the board, located just left of center of the 4-ft length and underneath where the driver is located (thus increasing the heat of the LEDs). Thermocouples will be placed at "G40," "G41," "G42," and on the driver TMPC (see **Figure C-1**).

The driver leads to the board are spliced to allow output power monitoring.

The DUT is reassembled, allowing for the thermocouples to exit the luminaire.

The DUT is to be tested per conditionally normative reference UL 1598, as Surface Mounted – Ceiling, per clause 19.10 (see **Section 6.1.3**).

The DUT is powered on and monitored for stabilization per **Section 7.1 DUT Thermal Stability** until power and temperatures are stabilized.

Once the DUT is stabilized, the measurements for T_A, T_C, $T_{S\text{-}G40}$, $T_{S\text{-}G41}$, and $T_{S\text{-}G42}$ are finalized. **Table C-1** shows examples of measured and normalized temperature values.

Table C-1. Measured and Normalized Temperature Values

Measured Value	Normalized Value
T_A = 28.4 °C	T_A = 25.0 °C
T_C = 60.6 °C	T_C = 57.2 °C
$T_{S\text{-}G40}$ = 51.9 °C	$T_{S\text{-}G40}$ = 48.5 °C
$T_{S\text{-}G41}$ = 52.2 °C	$T_{S\text{-}G41}$ = 48.8 °C
$T_{S\text{-}G42}$ = 51.6 °C	$T_{S\text{-}G42}$ = 48. 2°C

Drive current is determined by dividing the output current monitored from the driver to the LED by the number of parallel circuits (14 per schematic shown in **Figure C-2**). In this example, a measured current of 0.743 A is divided by 14, to give a reported value (drive current) of 0.0531 A.

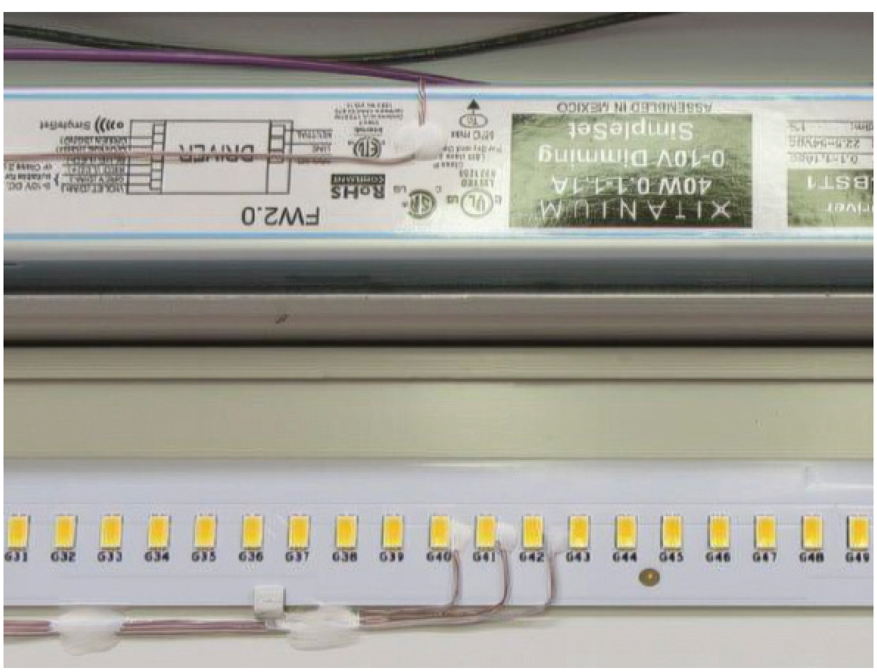

Figure C-1. Thermocouple placement on example LED board and driver.

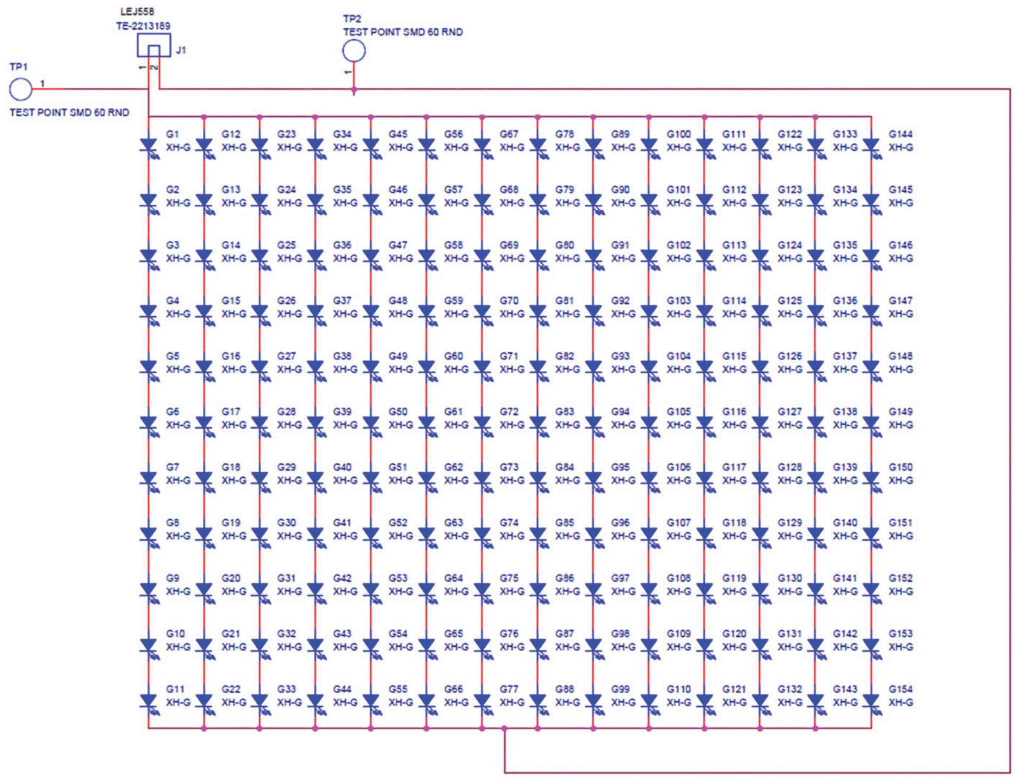

Figure C-2. SSL board schematic showing example parallel strings of LEDs.

C.2 LED Linear Ambient Luminaire Test Results and Uses

The corrected driver T_C (57.2 °C) can be used to validate the expected lifetime of the driver per the driver manufacturer's specification sheet.

Using the drive current for this example (0.0531 A), the maximum T_S (48.8 °C), and ANSI/IES LM-80 data for the SSL component, the ANSI/IES TM-21 calculations could be run to determine a lumen maintenance prediction for this luminaire.

INFORMATIVE REFERENCES

1 Illuminating Engineering Society. ANSI/IES LM-79-19, Approved Method: Optical and Electrical Measurements of Solid-State Lighting Products. New York: IES; 2019.

2 Illuminating Engineering Society. ANSI/IES TM-21-21, Technical Memorandum: Projecting Long-Term Lumen, Photon, and Radiant Flux Maintenance Of LED Light Sources. New York: IES; 2021.

3 International Bureau of Weights and Measures (BIPM). JCGM 100:2008, Guide to the Expression of Uncertainty in Measurement (GUM). Saint-Cloud, France: BIPM; 2008. Online: https://www.iso.org/sites/JCGM/GUM/JCGM100/C045315e-html/C045315e_FILES/MAIN_C045315e/Start_e.html. (Accessed 2024 Apr 1).

4 International Electrotechnical Commission (IEC). IECEE OD-5012, Laboratory Procedure for Acceptance, Preparation, Extension and Use of Thermocouples, Ed. 1.2 2021-01. Geneva, Switzerland: IEC; 2021.

Process for Change to an ANSI/IES Standard under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

a) Proposed change accepted for public review without modification
b) Proposed change accepted for public review with modification
c) Proposed change accepted for further study
d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE:
Use a separate form for each comment. Submit to the Director of Standards, IES, 85 Broad St, FL 17, New York NY 10004. Email: standards@ies.org. Fax: 212-248-5017.

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3. Clause (section), sub-clause or paragraph number; and page number: ______________________________
4. My proposal (check one):
   [ ] Change to read as follows
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Use underscore to show material to be added (<u>added</u>) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

____
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____

6. Reason and substantiation:

____
____
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Select as applicable:
   [ ] Additional pages are attached. Number of additional pages:
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Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:

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